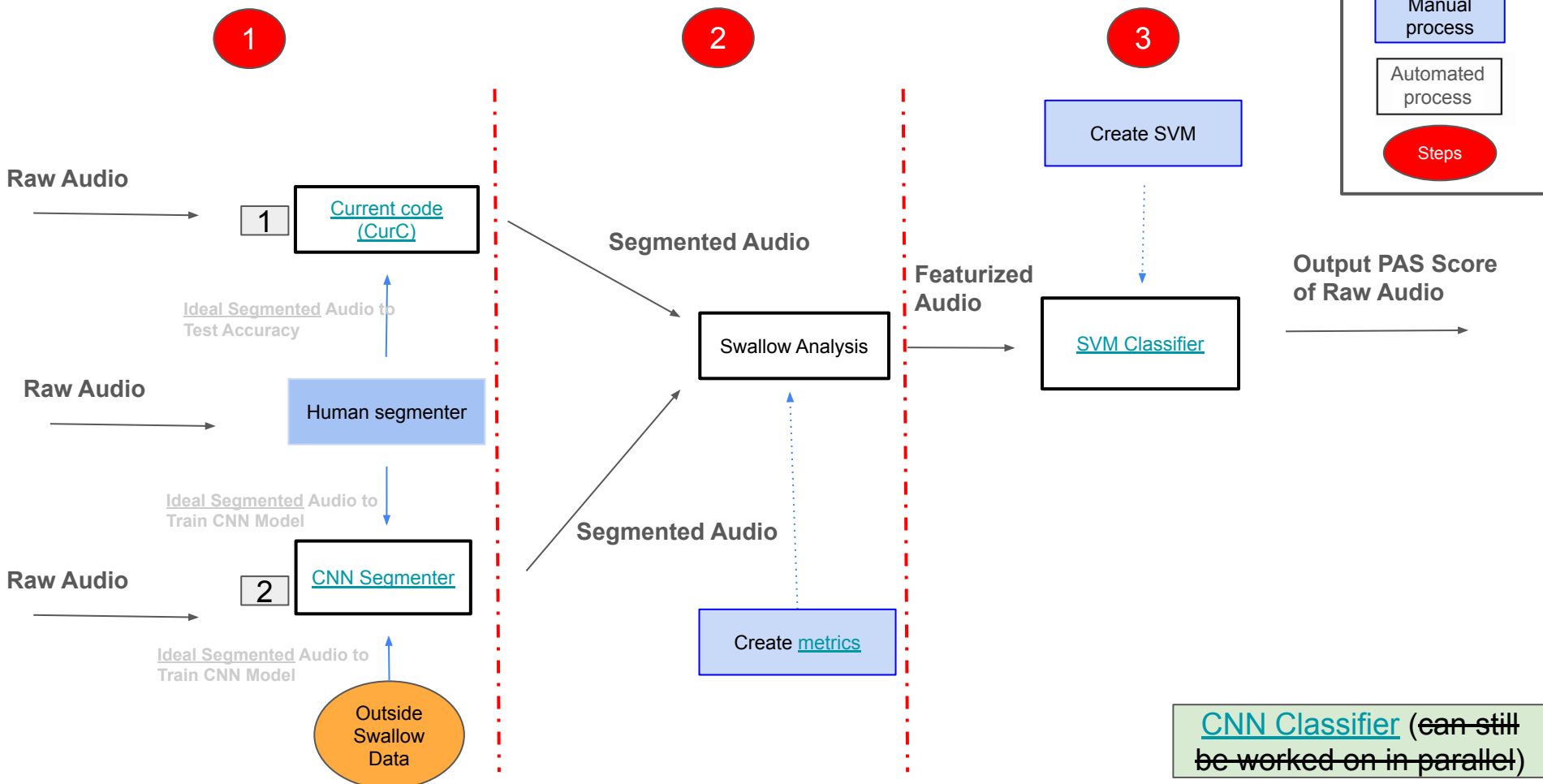
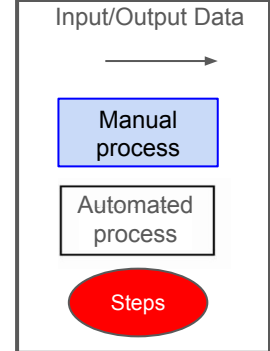


Swallowing Sound Analysis

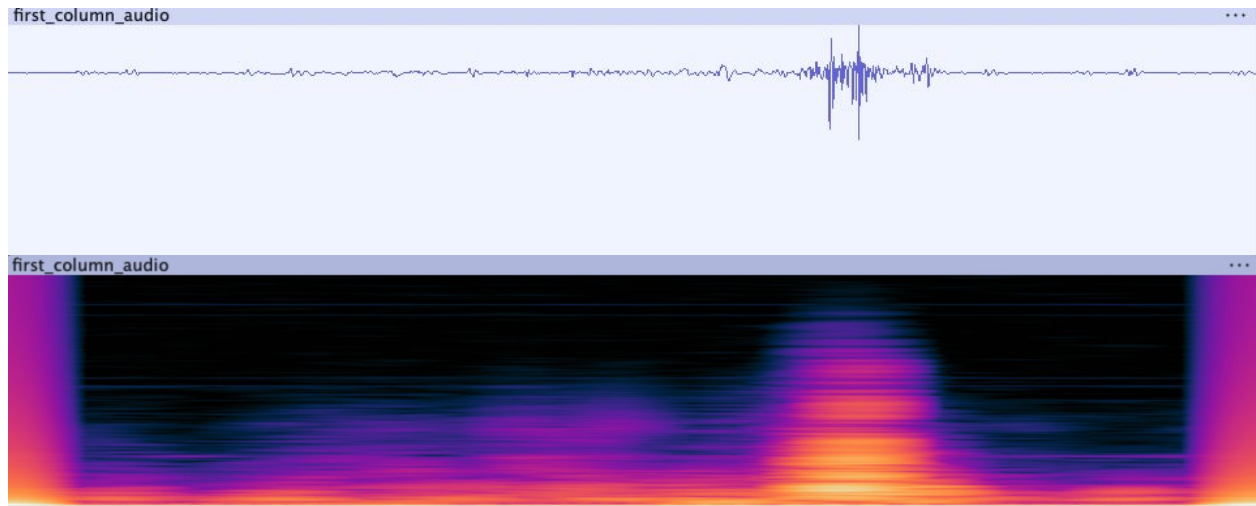
Yann Baglin-Bunod
Last update: 7/30/2024



Big Picture



This is what it should look like



Model Training

- Class for loading tensors and labels
- Dataloader from pytorch for batching
- GNet model with additional conv2d blocks to get input to 3x256x256
- Training code and saving model's final state
- Testing model accuracy on Semg dataset
- Testing model accuracy on Medical audio data
 - Medical data preprocessing

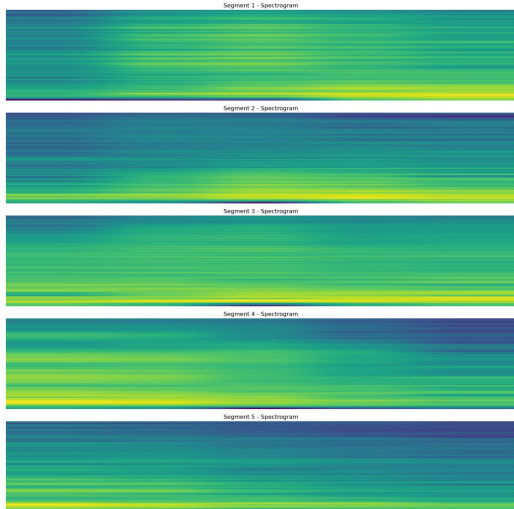
- ☐ Train the CNN Model:
 - ☐ Train the CNN model on the training data.
 - ☐ Validate the model on the validation set.
 - ☐ Evaluate the final model on the test set.
- ☐ Classify New Audio Files:
 - ☐ Preprocess new audio files by chunking them into 100ms segments and generating spectrograms.
 - ☐ Normalize the spectrograms to match the training data.
 - ☐ Use the trained model to classify each segment.
- ☐ Visualize Results:
 - ☐ Plot the waveform of the entire audio file.
 - ☐ Highlight the segments where the model predicted swallows.
 - ☐ Continue data augmentation techniques (add talking noise, and fix the gaussian noise)
🔗 next
 - ☐ Use the original dataset with train, test and validation
🔗 next
 - ☐ Use the smaller data snippets to see if classifier works

Interim step: Change of Spectrograms

- really bad → better but bad?

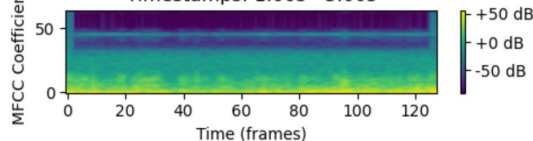
*these are non-augmented files

Original Spectrograms



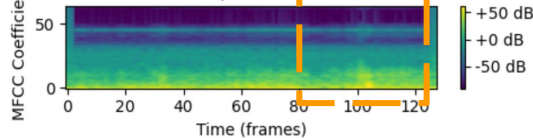
Labeled NON-SWALLOW

MFCC: 09_swallow_banana_segment_1_combined.pt
Segment Classification: 0
Timestamps: 1.00s - 3.00s



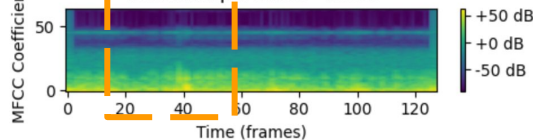
Label tensor values for 09_swallow_banana_segment_2_combined.pt: 0
(64, 128)

MFCC: 09_swallow_banana_segment_2_combined.pt
Segment Classification: 0
Timestamps: 2.00s - 4.00s



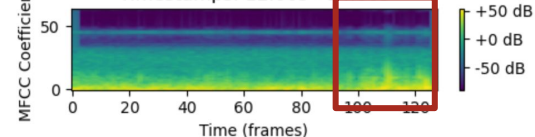
Label tensor values for 09_swallow_banana_segment_3_combined.pt: 0
(64, 128)

MFCC: 09_swallow_banana_segment_3_combined.pt
Segment Classification: 0
Timestamps: 3.00s - 5.00s



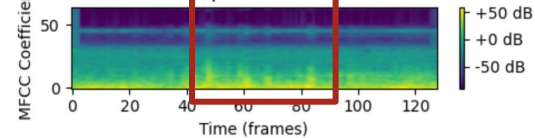
Labeled SWALLOW

MFCC: 09_swallow_banana_segment_12_combined.pt
Segment Classification: 1
Timestamps: 12.00s - 14.00s



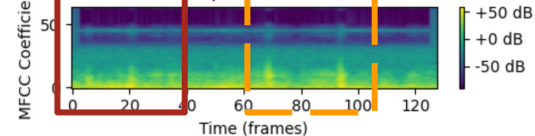
Label tensor values for 09_swallow_banana_segment_13_combined.pt: 1
(64, 128)

MFCC: 09_swallow_banana_segment_13_combined.pt
Segment Classification: 1
Timestamps: 13.00s - 15.00s



Label tensor values for 09_swallow_banana_segment_14_combined.pt: 1
(64, 128)

MFCC: 09_swallow_banana_segment_14_combined.pt
Segment Classification: 1
Timestamps: 14.00s - 16.00s



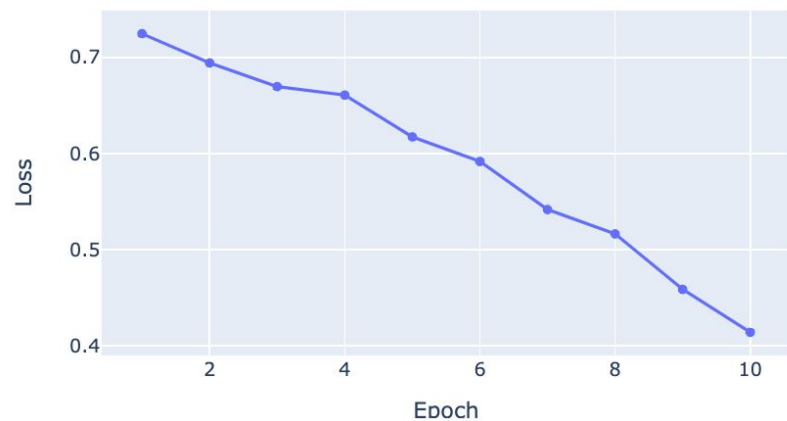
Previous Action Step

- Finalize model training and see results

- 0 accuracy

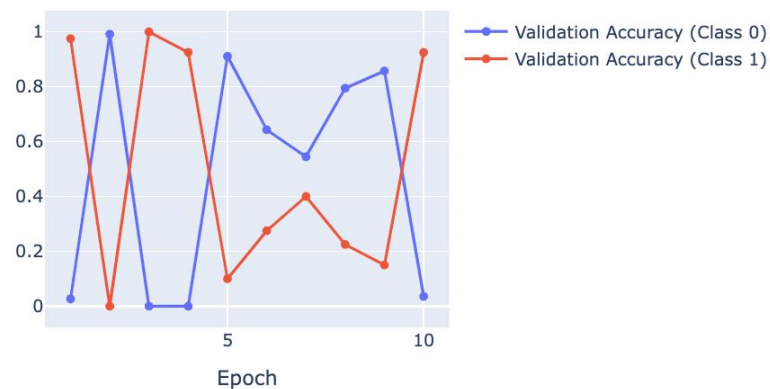


Training Loss - lr=0.001, momentum=0.9, weight_decay=0.0



With just original files: 789 0-labeled, 226 1-labeled spectrograms

Validation Accuracy - lr=0.001, momentum=0.9, weight_decay=0.0



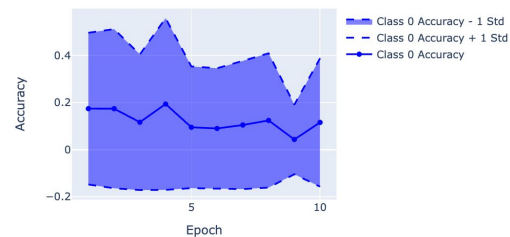
Number of labels as 0: 9456 (8667 increase)
Number of labels as 1: 2724 (2498 increase)

Using Data Augmented Files

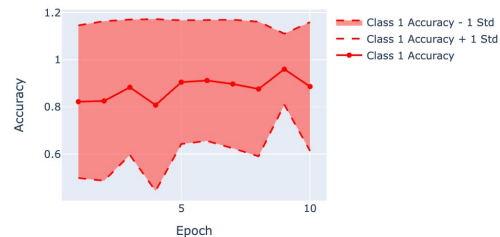
- Large number difference between classes ~4:1
 - Higher penalty for misclassification of minority class
 - `class_weights = torch.tensor([1.0, 5.0], dtype=torch.float)`
 - `criterion =`
`nn.CrossEntropyLoss(weight=class_weights.to(device))`
- Outcome: same issue
 - CNN model learns to almost always classify image as a swallows

Average accuracy by class over 75 trainings

Validation Accuracy for Class 0

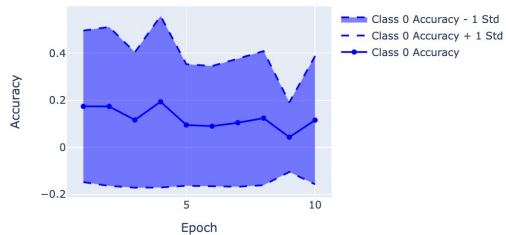


Validation Accuracy for Class 1

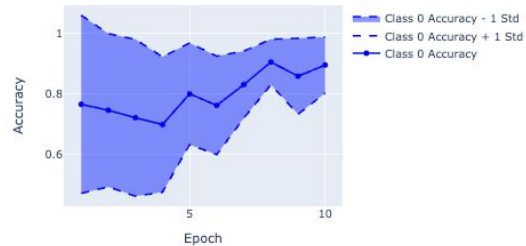


Validation accuracy updated

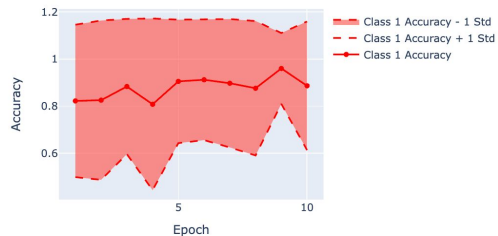
Validation Accuracy for Class 0



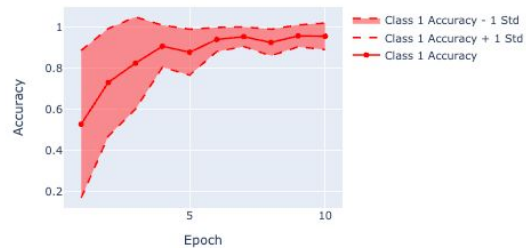
Validation Accuracy for Class 0



Validation Accuracy for Class 1

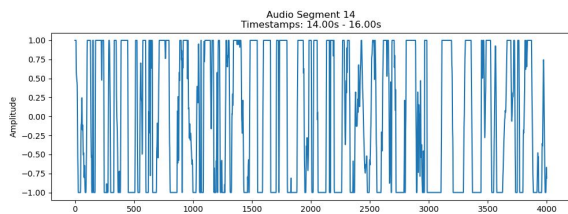


Validation Accuracy for Class 1

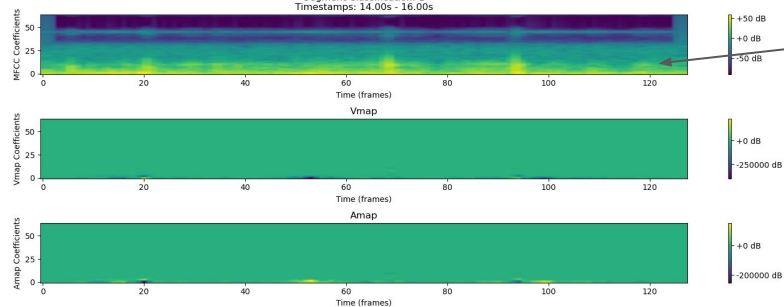
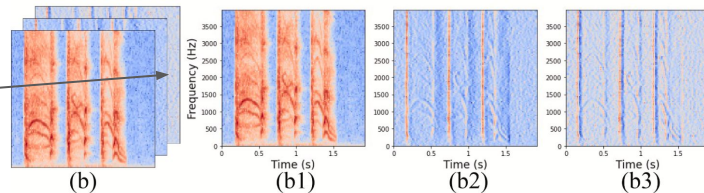


Comparison to Paper

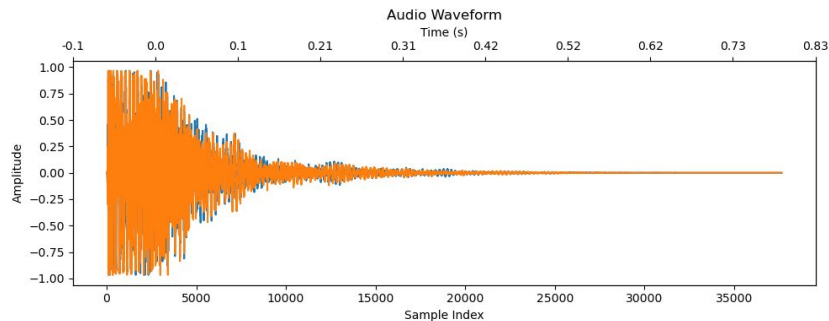
- Model spectrograms are different from the ones that I am using.
 - Not easily comparable (no access to dataset used)



128x128 image
of spectrogram



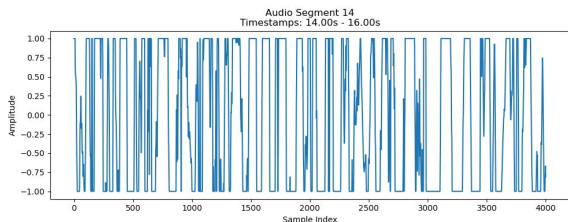
64x128 image
of spectrogram



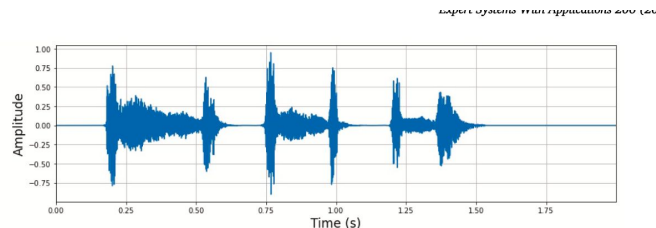
Sample Rate Difference

- ASR standard: speech is bwn 512Hz and 2,048Hz
 - Means that the sampling rate must 2x higher - ~4kHz (nyquist sampling)
 - Standard is usually 8kHz or higher
- Swallowing sound is between 0–2 kHz (Dry swallow: 1,000–1,100 Hz. Wet swallow: 400–1000 Hz)
 - Means sampling rate must be ~4kHz

2sec 2kHz sample



2sec 16kHz sample



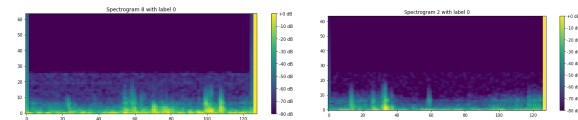
Problems?

Possible issues:

- Data Doesn't Capture Swallows
- Not Enough Data

Conclusion

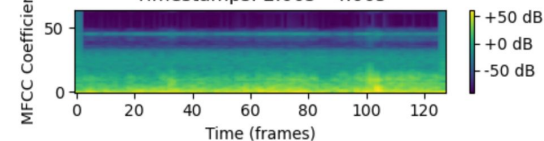
- Too small dataset!
 1. “Noise” can’t be taken out
- Possible actions
 - Find new datasets
 - PubMed: 0 (searched ~20 related articles)
 - Create a better Spectrogram representation



MFCC: 09_swallow_banana_segment_2_combined.pt

Segment Classification: 0

Timestamps: 2.00s - 4.00s

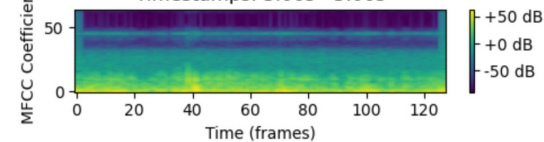


Label tensor values for 09_swallow_banana_segment_3_combined.pt: 0
(64, 128)

MFCC: 09_swallow_banana_segment_3_combined.pt

Segment Classification: 0

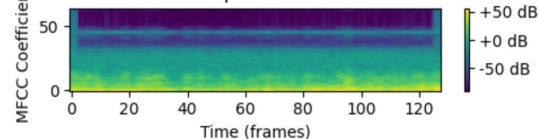
Timestamps: 3.00s - 5.00s



MFCC: 09_swallow_banana_segment_1_combined.pt

Segment Classification: 0

Timestamps: 1.00s - 3.00s



0 labeled
“Noisy
Images”

0 labeled
“non-noisy”

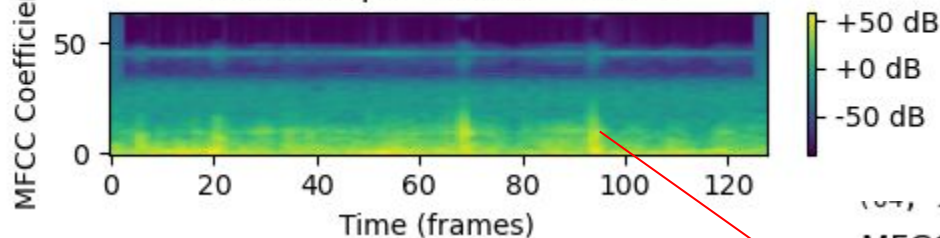
Addressing issues

- Adding a hamming window to reduce the edges' importance
 - helps to minimize spectral leakage, resulting in a clearer and more accurate representation of

MFCC: 09_swallow_banana_segment_14_combined.pt

Segment Classification: 1

Timestamps: 14.00s - 16.00s



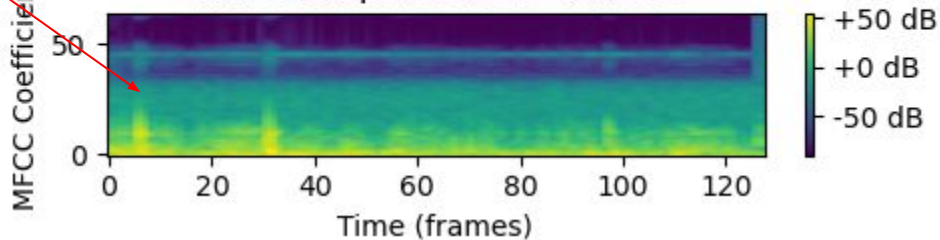
Less emphasis on repeat information that could be "noise"

(0 to 1) chance in class but still same spectral parts

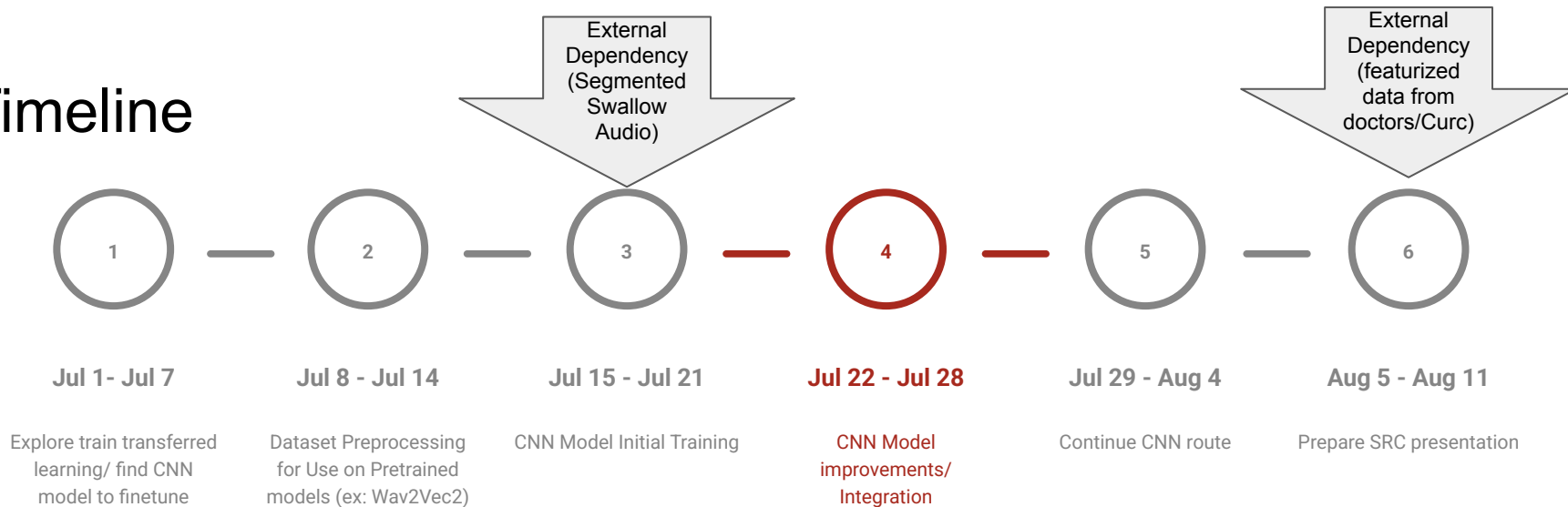
MFCC: 09_swallow_banana_segment_15_combined.pt

Segment Classification: 0

Timestamps: 15.00s - 17.00s



Timeline



SRIP
Important
Dates

JULY 7TH	STUDENT PRESENTER REGISTRATION AND ABSTRACT DEADLINE
STUDENT PRESENTER MODIFICATION DEADLINE	JULY 14TH
AUGUST 9TH	"HOW TO PRESENT AT THE SRC" ZOOM PRESENTATION - 9AM
SUMMER RESEARCH CONFERENCE! PRICE CENTER	AUGUST 14TH & 15TH



Creating Own Model for Task

- Base off of research paper: Deep learning based cough detection using enhanced features
- Much more focused on swallow data
 - Searching for a pretrained model can be done if this model doesn't work
- Can be used to be fine tuned to coughs

Data Augmentation

- Background gaussian noise
- Audio shift
- Audio speed change
- Background talking noise

Increased initial dataset from 196 seconds of swallow audio to train on to ~20 minutes.

Dataset Contents

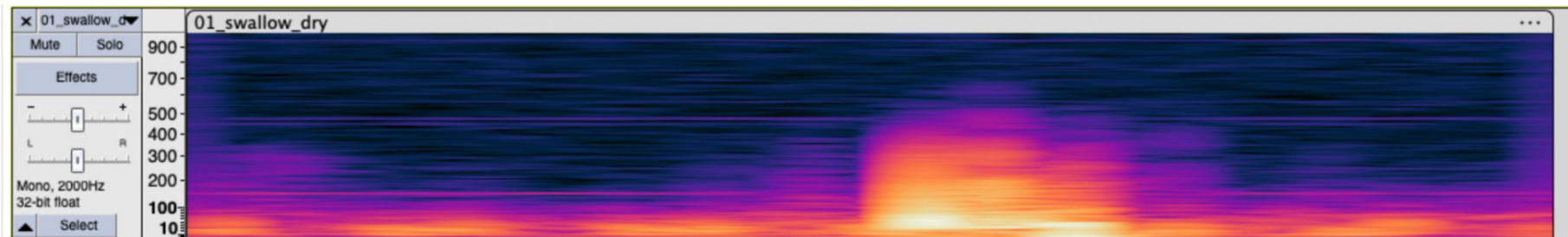
Semg total amount of seconds according to labels:

- Total seconds with label None: 995.947
- Total seconds with label Swallow: 179.7405

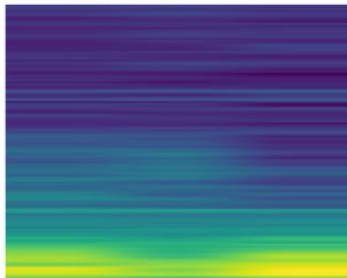
Audio Segmentation and Feature Extraction

Problems with: n_{fft} size, window size, and segment size

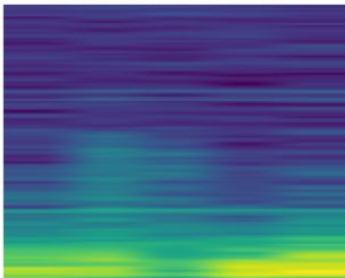
- Goal: create small windows of audio to train model



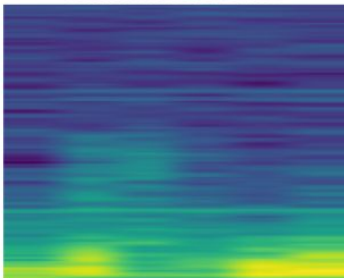
Segment 1 - FFT: 2048 - Label: 0
Window: 1.024s, Hop: 0.512s
Dimensions: 128 x 3



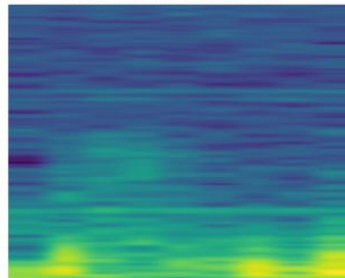
Segment 1 - FFT: 1024 - Label: 0
Window: 0.512s, Hop: 0.256s
Dimensions: 128 x 5



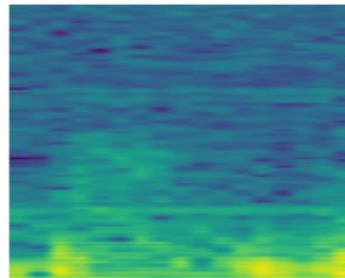
Segment 1 - FFT: 768 - Label: 0
Window: 0.384s, Hop: 0.192s
Dimensions: 128 x 6



Segment 1 - FFT: 512 - Label: 0
Window: 0.256s, Hop: 0.128s
Dimensions: 128 x 9



Segment 1 - FFT: 256 - Label: 0
Window: 0.128s, Hop: 0.064s
Dimensions: 128 x 17



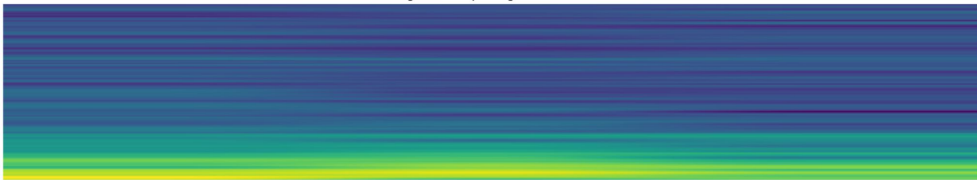
Audio Segmentation and Feature Extraction

- Chose 250ms snippets with overlap
- Created V-maps and A-maps for each audio
- Final output:
 - **3x128x128 matrices for model training**

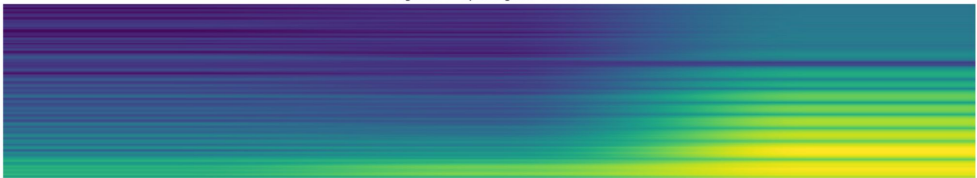
$$V(t,f) = \begin{cases} X(t+1,f) - X(t,f); (t=0), \\ \frac{X(t+1,f) - X(t-1,f)}{2}; (1 \leq t \leq T-1), \\ X(t,f) - X(t-1,f); (t=T), \end{cases} \quad (3)$$

$$A(t,f) = \begin{cases} V(t+1,f) - V(t,f); (t=0), \\ \frac{V(t+1,f) - V(t-1,f)}{2}; (1 \leq t \leq T-1), \\ V(t,f) - V(t-1,f); (t=T). \end{cases} \quad (4)$$

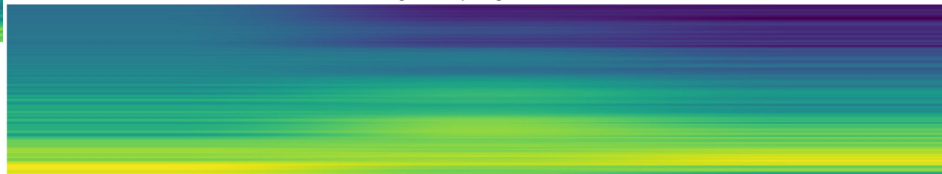
Segment 3 - Spectrogram - Label: 0



Segment 7 - Spectrogram - Label: 1



Segment 8 - Spectrogram - Label: 1



Abstract

Dysphagia, or difficulty swallowing, is a condition that can result from various medical disorders, and its assessment typically requires expensive and invasive methods such as endoscopy. This study aims to develop an automated acoustic analysis system to classify swallow capabilities based on audio recordings, offering a non-invasive and cost-effective alternative. The research involves collecting audio data during swallowing events, segmenting the audio to isolate swallows, and extracting relevant features using Mel-frequency cepstral coefficients (MFCCs), V-maps, and A-maps. A Convolutional Neural Network (CNN) is employed to process the audio and segment the swallowing events. An SVM is employed to classify the swallowing events according to the Penetration-Aspiration Scale (PAS), which ranges from 1-8. Preliminary results demonstrate a differentiation between normal and abnormal swallows, suggesting the potential for this method to provide accurate and reliable assessments of swallowing function. Future work includes enhancing the dataset through data augmentation techniques, refining the CNN model for improved accuracy, and validating the system in clinical settings.

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(updated 6/27/2024)

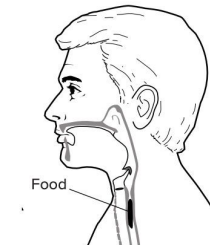
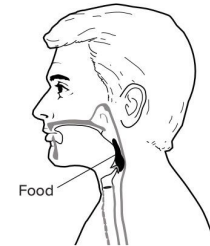
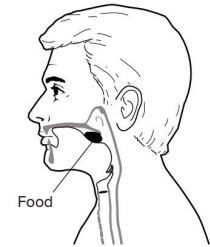
List of papers

Ref paper that does same as us: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7349358/>

Paper to base model off of: <https://www.sciencedirect.com/science/article/pii/S0957417422010740>

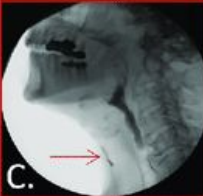
Background

- Conditions can cause patients not to be able to swallow normally.
- Only extremely effective way to assess someone's swallow capabilities is expensive.
 - \$\$\$ Camera to examine swallow (endoscope)
 - If not done with this, there is a discernible error in diagnosing patients.



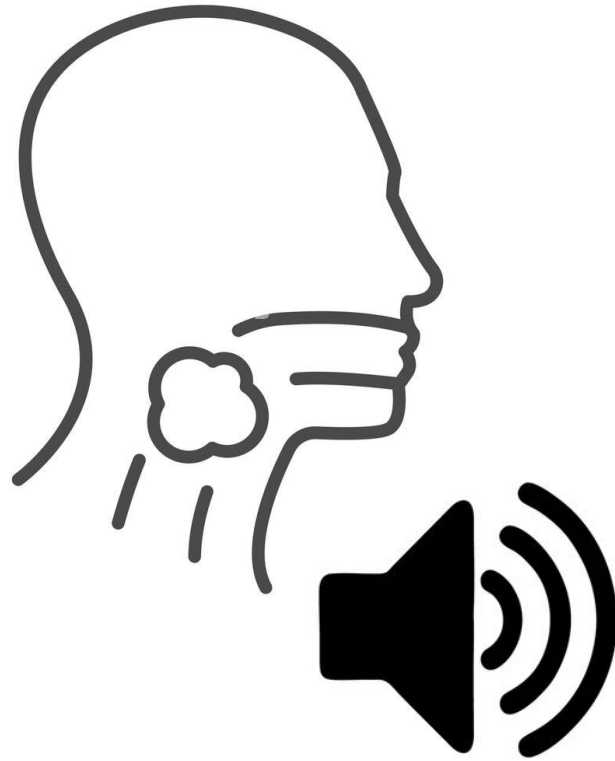
Scale

- Penetration-aspiration scale (PAS) - used to characterize both the location of airway invasion events and a patient's response during videofluoroscopic swallowing studies
 - PAS 1-2 is normal
 - PAS 3 - 8 abnormal

PAS Score Description:		Safety Status:	
1	Material does not enter the airway	Safe	 A.
2	Material enters the airway, remains above the vocal folds, is ejected from airway		
3	Material enters the airway, remains above vocal folds, is not ejected from airway	Unsafe	 B.
4	Material enters the airway, contacts the vocal folds, is ejected from airway		
5	Material enters the airway, contacts the vocal folds, is not ejected from airway		
6	Material enters the airway, passes below vocal folds, ejected from trachea		
7	Material enters the airway, passes below vocal folds, not ejected despite effort		
8	Material enters the airway, passes below the vocal folds, no effort made to eject		 C.

Is There a Better Way?

- Sound Recording of Swallows



Sound Recording Protocol

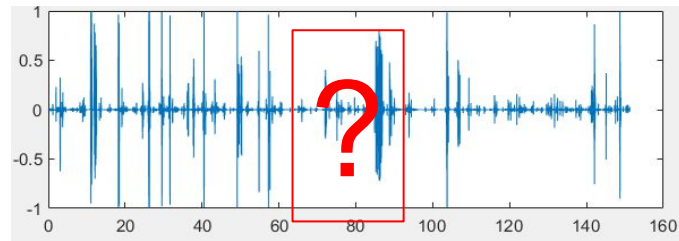
- **Preparation:** Obtain patient consent and position a stethoscope lateral to the thyroid cartilage before FEES.
- **Feeding Protocol:** Follow a consistent order of thin liquids, semi-solids, and solids; begin recording at clinician's signal.
- **Procedure Management:** No counseling during FEES; if over 2 minutes, pause for a 10-second stethoscope reset.

Objective

Demonstrate clear differentiation between different swallow capabilities from recordings.

Goals:

1. Analyze swallow data
 - a. Derive proof of differentiations between labels (swallow capabilities)
2. **Create accurate machine learning algorithm for classification.**

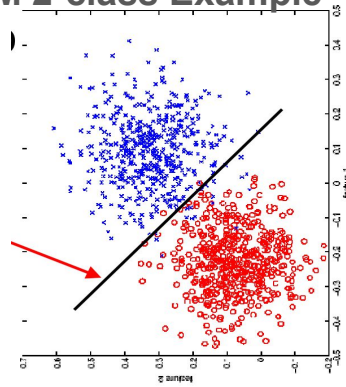


Segmenter/SVM

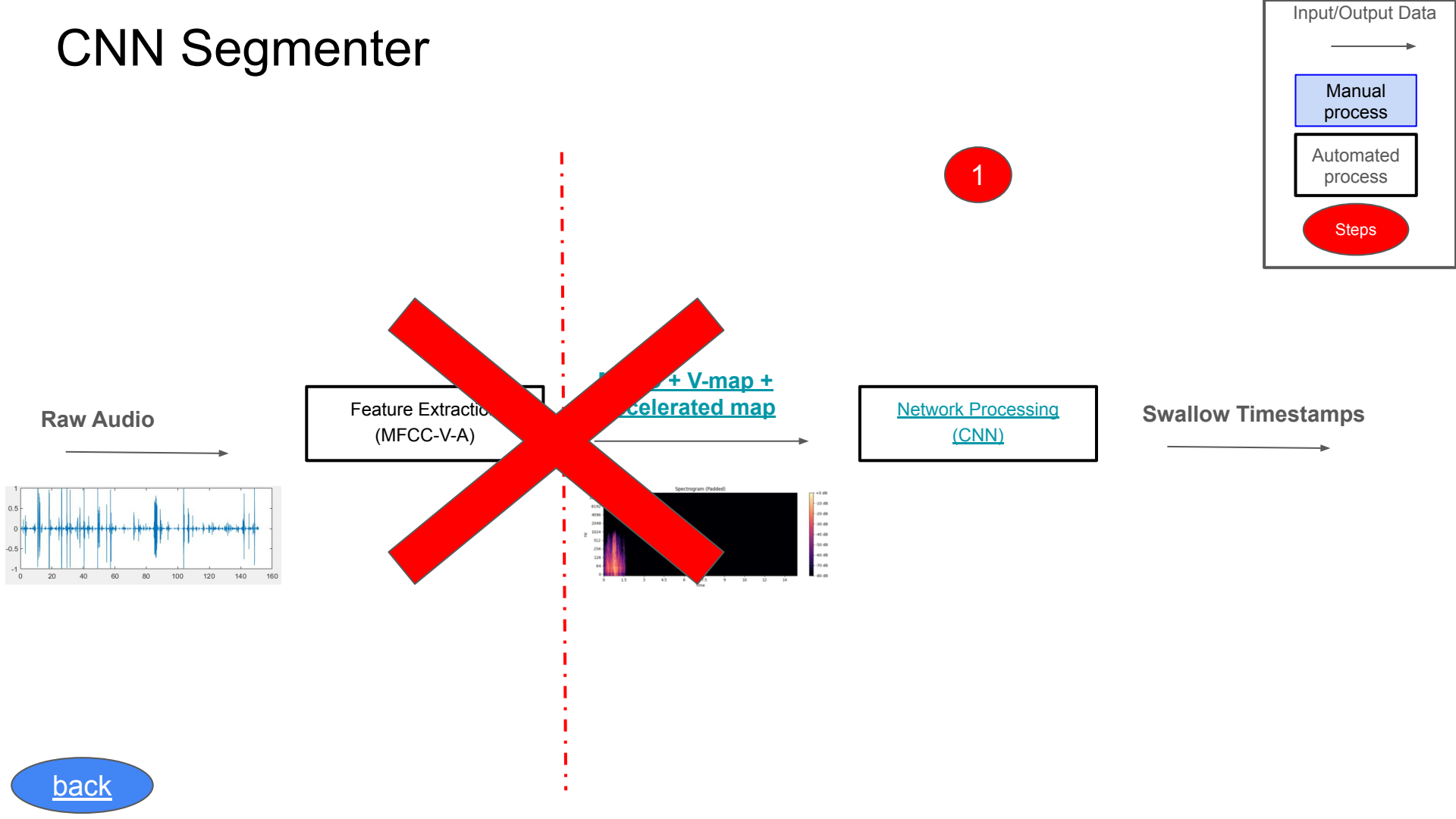
Summary

- Proposed idea is to create an automatic classifying system on patients from audio produced.
- For the Features to SVM classification system: the only way to really improve accuracy is to create good data and choose the right features*. Same as CNN previously
 - Current labelling system still necessary for SVM as they were for the CNN classification
 - Now we just need to find a way to segment the audio data

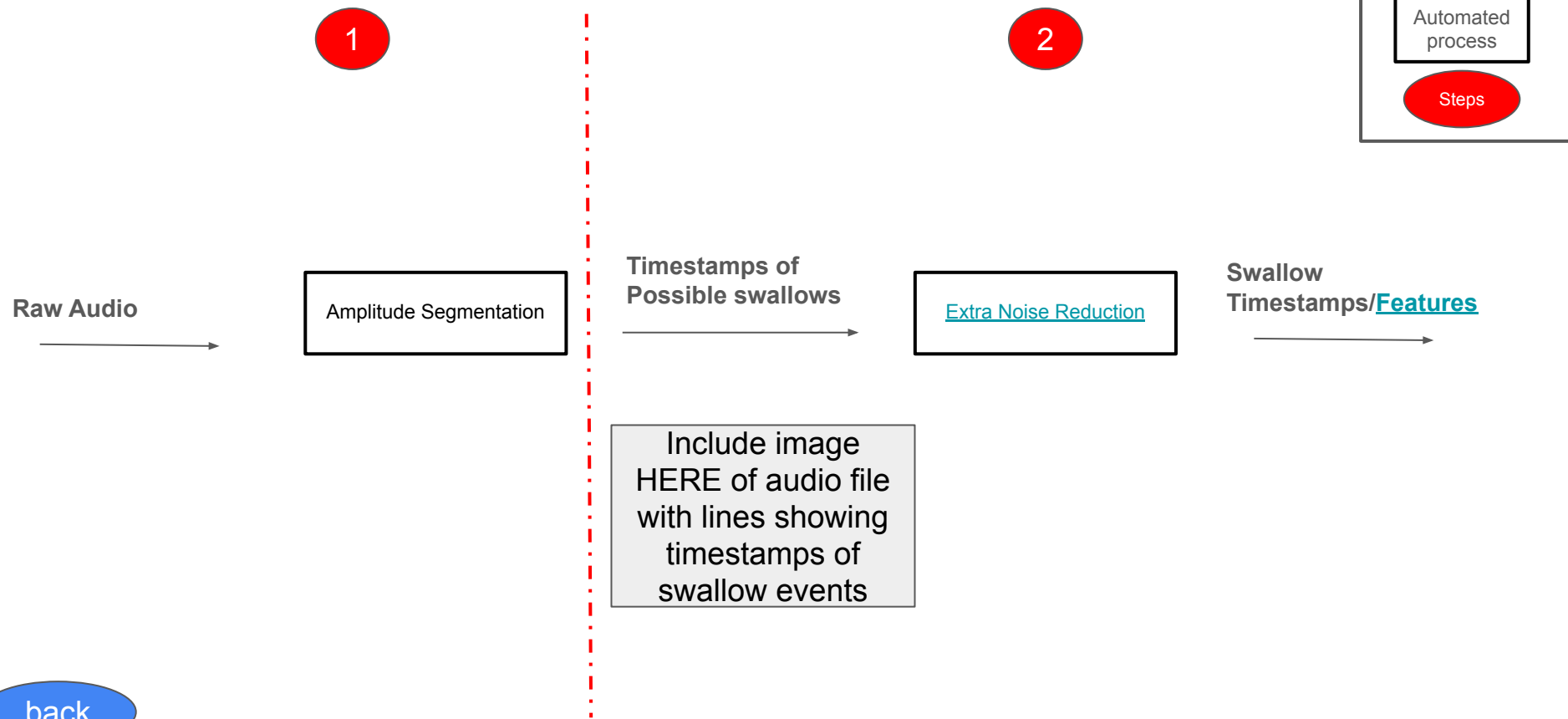
SVM 2-class Example



CNN Segmenter



Current Code



Extra Noise Reduction

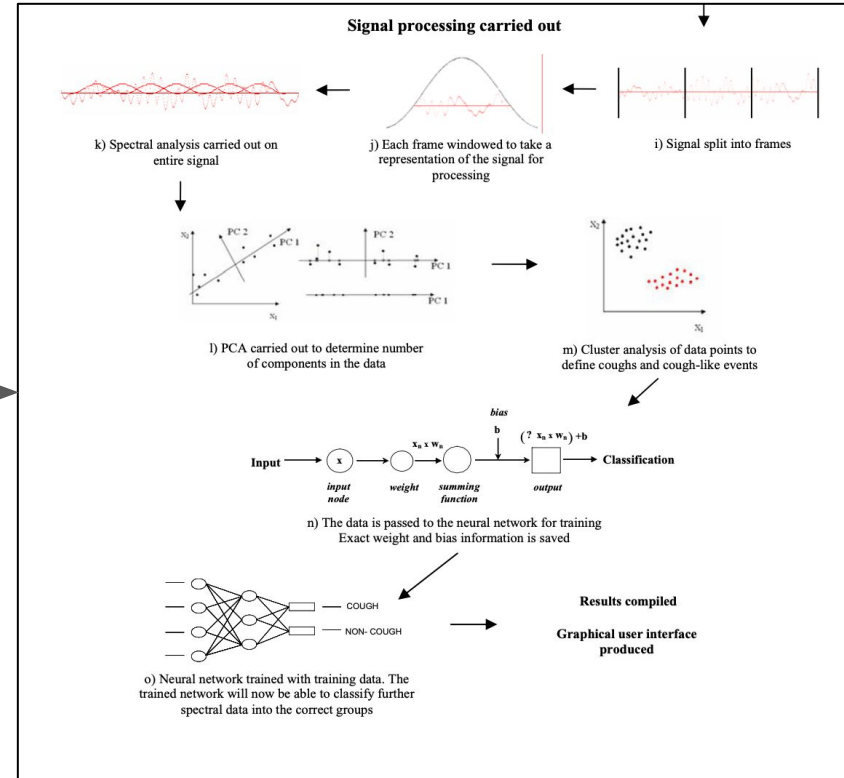
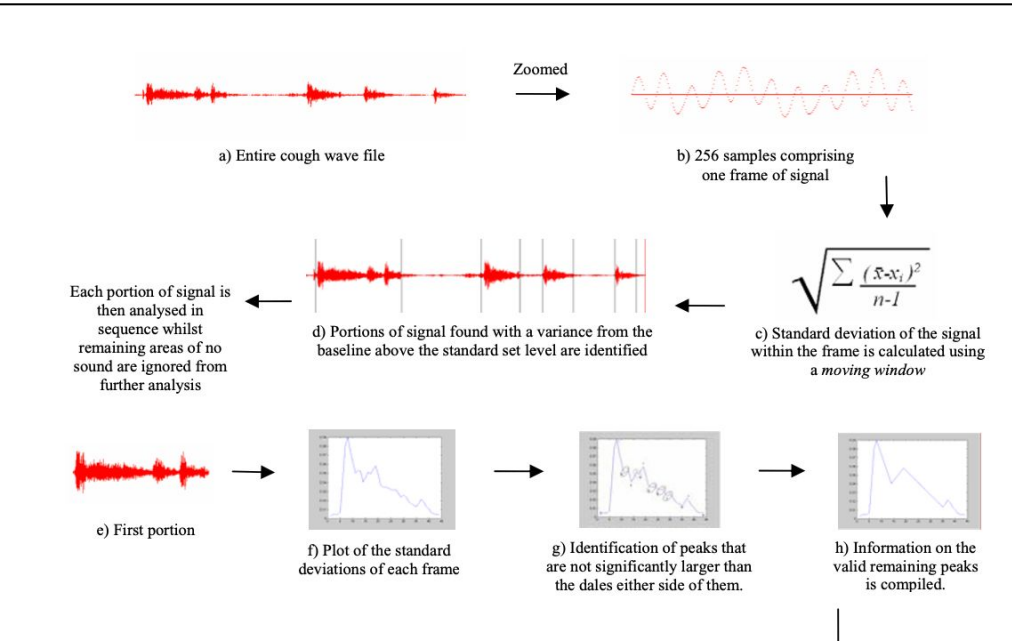
So far:

- limit loud sounds (“aaaaa”, “eeee”...)
- Combine close timestamps into one group

Paper to look at: [The automatic recognition and counting of coughs](#)

- In this paper, they speak of ways to more easily find the coughs found in a very long audio file for manual classification (1 hour long).
 - Uses
 - That manual classification is then sent to a Neural Network for training!

The automatic recognition and counting of coughs



Ideal Segmented Swallow Audio

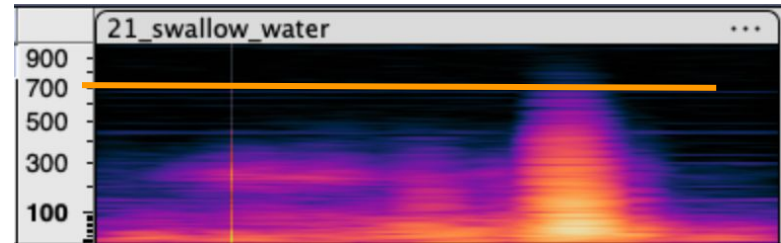
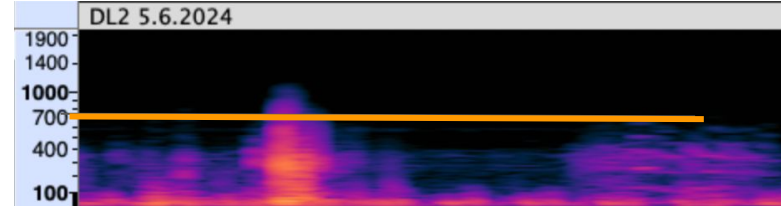
- Take output of analyzed swallows from Andrew
- Outside data
 - **Similar sound, also microphone placed on throat**
 - sEMG swallow/cough dataset [link](#)

Total number of Audio Files:

Total Swallow: 765

Total Cough: 132

[back](#)



Swallow metrics

1. PAS score of sample
2. Total Length (s)
3. Number of Swallows
4. Swallow # Length (s)
5. Num Swallows per Bolus
6. Num Coughs
7. Peak Amplitude
8. Average Amplitude
9. Amplitude²
10. (Amplitude²) per sec
11. Area under curve
12. Area under curve per second

Network Processing

1. Train Transfer Learning

- Find binary classifier model that I can finetune on data

~~2. Model from scratch~~

- G-net is best cough classifier in paper

PS: NEED Data Augmentation

[back](#)

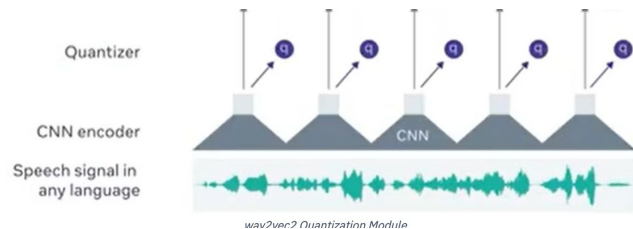
Table 4

G-net configuration for binary classification.

Type	Kernel size / Stride	Output size (C × H × W)	#1 × 1	#3 × 3 reduce	#3 × 3	#5 × 5 reduce	#5 × 5	Pool
Conv + GroupNorm + ReLU	7 × 7 / 1	16 × 128 × 128						
MaxPool	3 × 3 / 2	16 × 64 × 64						
Conv + GroupNorm + ReLU	3 × 3 / 1	48 × 64 × 64			48			
MaxPool	3 × 3 / 2	48 × 32 × 32						
Inception Module 1		256 × 32 × 32	64	96	128	16	32	32
Inception Module 2		480 × 32 × 32	128	128	192	32	96	64
MaxPool	3 × 3 / 2	480 × 16 × 16						
Inception Module 3		512 × 16 × 16	192	96	208	16	48	64
Inception Module 4		512 × 16 × 16	160	112	224	24	64	64
Inception Module 5		512 × 16 × 16	128	128	256	24	64	64
Inception Module 6		528 × 16 × 16	112	144	288	32	64	64
Inception Module 7		832 × 16 × 16	256	160	320	32	128	128
MaxPool	3 × 3 / 2	832 × 8 × 8						
Inception Module 8		832 × 8 × 8	256	160	320	32	128	128
Inception Module 9		1024 × 8 × 8	384	192	384	48	128	128
AvgPool	8 × 8 / 1	1024 × 1 × 1						
FullyConnected		2 × 1 × 1						
Softmax		2 × 1 × 1						

Train-Transfer Learning

- Wav2Vec2 pretrained model:
 - Pretrained on 57k hours of unlabelled audio data
 - Learns to find audio units (speech and other units)
 - [Wav2Vec2-BERT](#): trained on 4.5M hours
 - 580M parameters



- Tasks
 - Preprocessing [Datasets](#) for Swallow and Cough
 - Fine Tuning using Pytorch

Preprocessing

Files will be:

1. Audio as numpy array
2. Label with values of 0 or 1 depending on if it is a swallow segment or not

Data Augmentation

- **Since there is a data imbalance we need to synthically create more data for classes 2-8**
 - Link to SMOTE technique: <https://arxiv.org/abs/1106.1813>

MFCC+V-Map+A-Map

- MFCC: Mel-scale of the cepstrum (fast fourier transformed signal; FFT)
 - Mel-scale approximates the response of the human auditory system (it focuses on frequency shape humans produce and attempts to accurately represent it)
- V-map, A-map
 - V-map represents the amount of change in a spectrogram-based feature over time
 - A-map represents the amount of change in a V-map over time

$$V(t,f) = \begin{cases} X(t+1,f) - X(t,f); (t=0), \\ \frac{X(t+1,f) - X(t-1,f)}{2}; (1 \leq t \leq T-1), \\ X(t,f) - X(t-1,f); (t=T), \end{cases}$$

$$A(t,f) = \begin{cases} V(t+1,f) - V(t,f); (t=0), \\ \frac{V(t+1,f) - V(t-1,f)}{2}; (1 \leq t \leq T-1), \\ V(t,f) - V(t-1,f); (t=T). \end{cases}$$

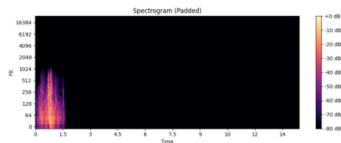
Ref:

<https://www.sciencedirect.com/science/article/pii/S0957417422010740>

SVM Classifier

SVM Classifier

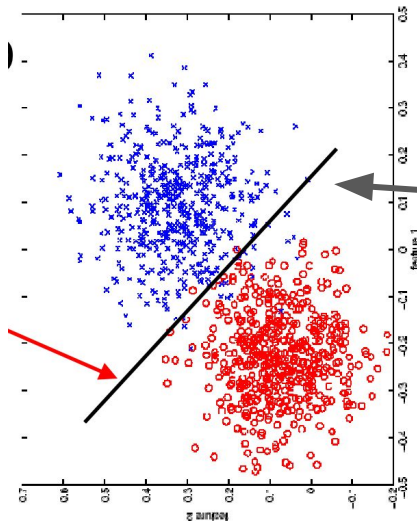
1. CNN



Classify scores using
learned model

2. SVM

2-class case



Classifying scores using
learned hyperplane

$$w^T x + b = 0$$

Steps Towards Using SVM

1. CNN Segmenter

Research Papers:
Ubiquoustics/ Flusense

Summary:
employ trained transfer
learning techniques on
pretrained audio
classifier model

Output:
Extracts swallows from
audio

2. Create Metrics for Swallow Audio

Use features outlined
by analysis project

Find other features
(frequency analysis!)

Total feature count
currently: **14**

3. Support Vector Machine

Vectorize Features

Use scikit-learn library
to visualize features
and then classify

Support Vector Machine - using a linear discriminant

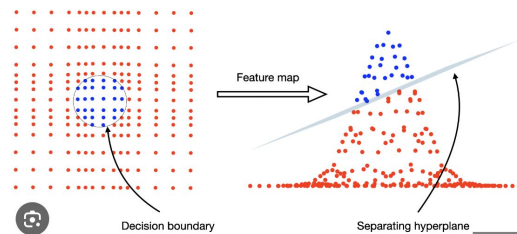
- Linear discriminant classifier that maximizes margin subject to constraints

1. **Create feature vectors for each Audio file**

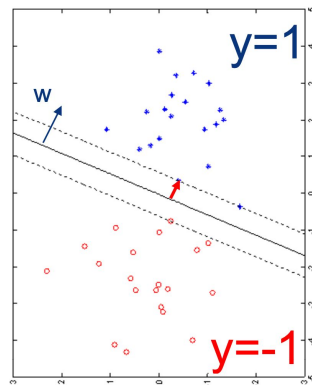
2. **Scikit-learn library!**

a. Perform SVM (essentially loss fn: tries to fit hyperplane to data)

i. *Kernelization necessary if problem is un-separable

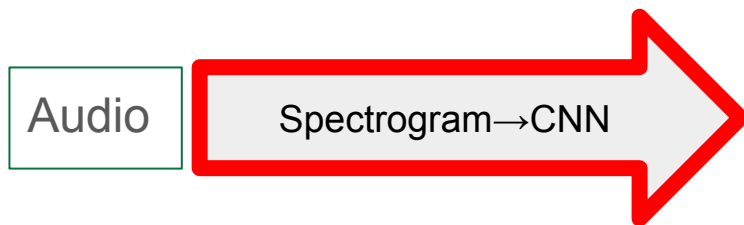


2-class case:



CNN Classification

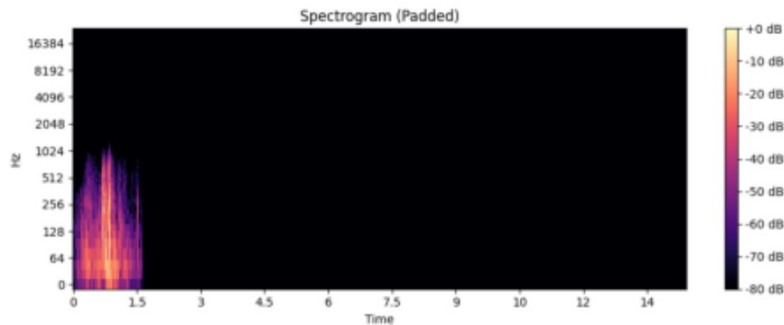
Audio ML



PAS Score Description:		Safety Status:	
1	Material does not enter the airway	Safe	A.
2	Material enters the airway, remains above the vocal folds, is ejected from airway		
3	Material enters the airway, remains above vocal folds, is not ejected from airway	Unsafe	B.
4	Material enters the airway, contacts the vocal folds, is ejected from airway		
5	Material enters the airway, contacts the vocal folds, is not ejected from airway		C.
6	Material enters the airway, passes below vocal folds, ejected from trachea		
7	Material enters the airway, passes below vocal folds, not ejected despite effort		
8	Material enters the airway, passes below the vocal folds, no effort made to eject		

CNN structures to try: VGGnet, Resnet, Alexnet

Dataset Transformation for CNN



CC 4.9.2024 Applesauce1_1_2_2_1.png

overall PAS

yale
residue in
vallecula

yale residue in
pyriform

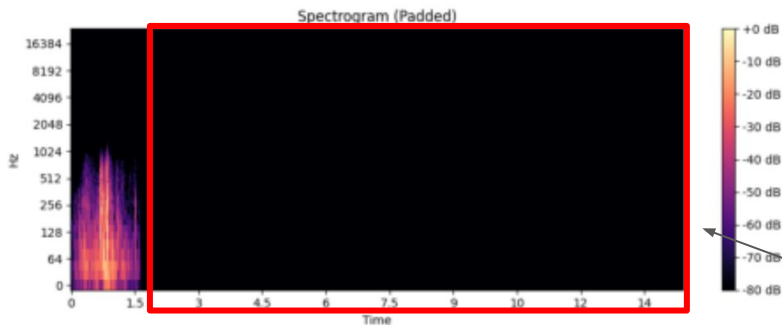
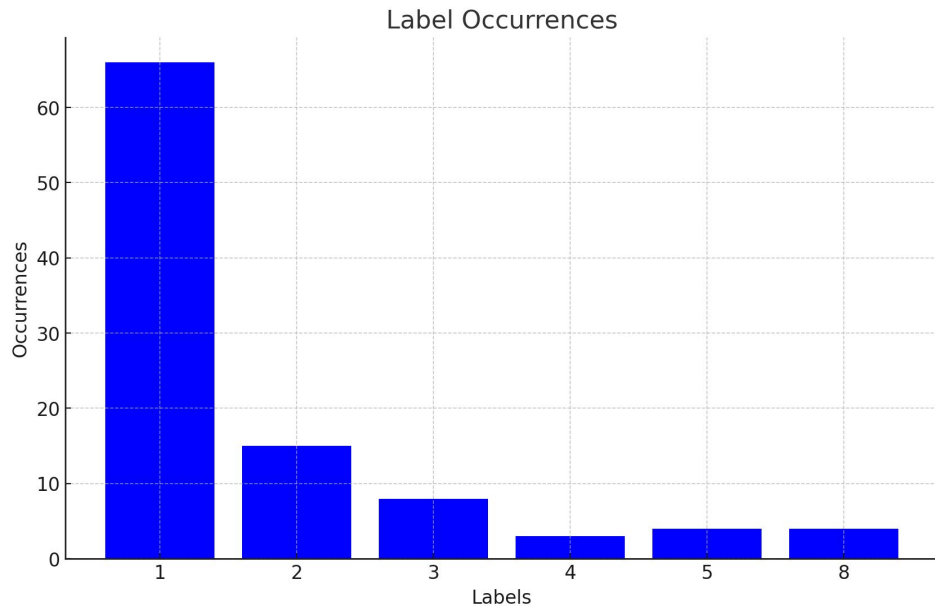
PAS of that trial

Current dataset: last updated 6/8/24

- Current Patient Count: 12
- Current individual Swallow samples: 101

Preprocessing

- Zero Padding
 - Large difference between max and min audio length
- Imbalance of dataset
 - More PAS 1 labels than other labels
 - Current Unique labels: 1,2,3,4,5,8
 - Missing 6,7
 - Data Augmentation
- Noise Cancellation



Zero Padding

Simple CNN

- Current model accuracy ranges from ~50%
 - Preprocessing through data augmentation and a change in zero padding can help

```
SimpleCNN(  
  (model): Sequential(  
    (0): Conv2d(4, 32, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1))  
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (2): ReLU()  
    (3): Conv2d(32, 64, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1))  
    (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (5): ReLU()  
    (6): Conv2d(64, 128, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1))  
    (7): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)  
    (8): ReLU()  
    (9): AdaptiveAvgPool2d(output_size=(1, 1))  
    (10): Flatten(start_dim=1, end_dim=-1)  
    (11): Linear(in_features=128, out_features=6, bias=True)  
  )  
)
```

Improvements to CNN

- Centering of data in padding (equal padding on either side)
- Data augmentation to address imbalance
 - Noise injection
 - Change speed/pitch
- Problem with time: Spectrograms may not capture this?
 - A shorter swallow is person who can swallow well
 - A much longer swallow is person who can swallow less well
- Questions
 - Asking people to say “eeee” to make the audio could be helpful?

Approximate Timeline

Ablation Study

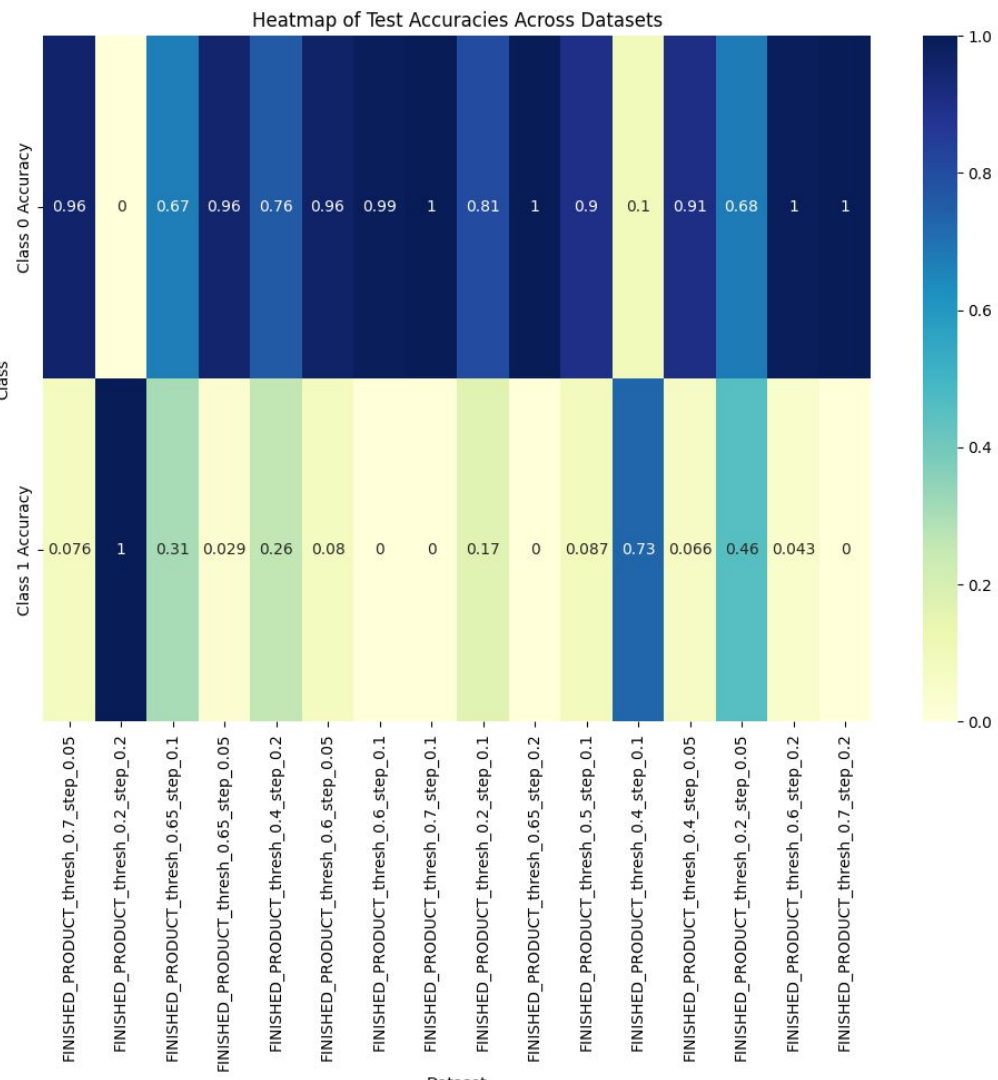
Chose to change:

label_thresholds = [0.2, 0.4, 0.6, 0.65, 0.7]

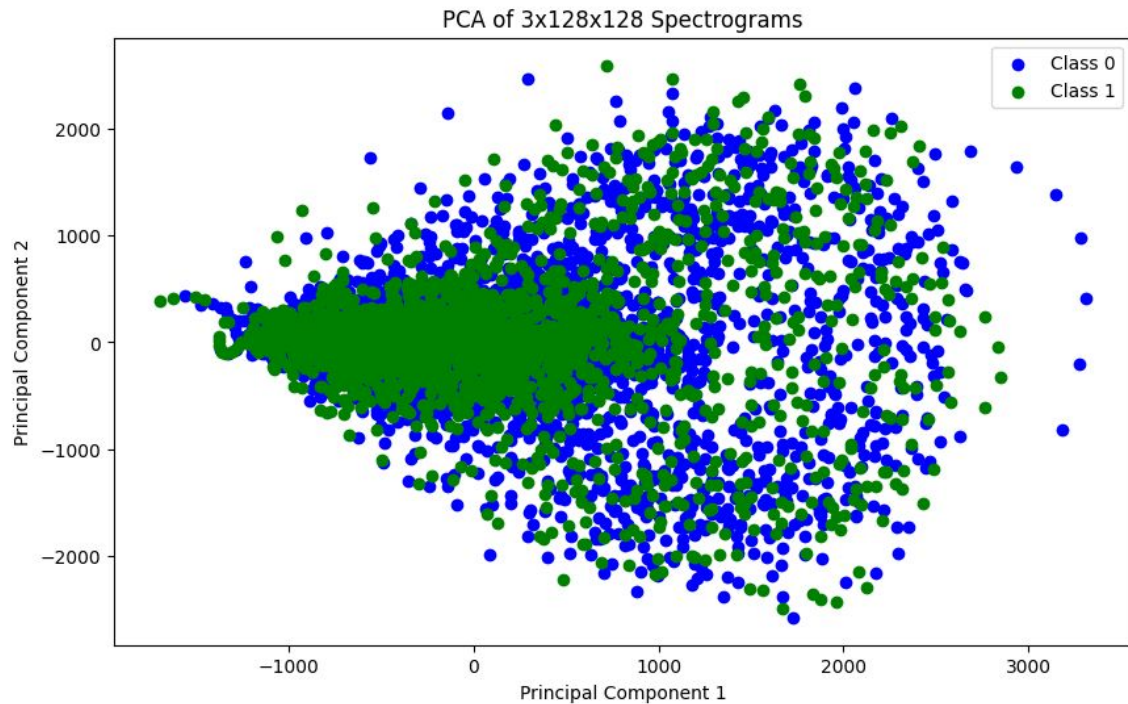
- min # of 1s to for label segment to be labeled 1

step_sizes_in_sec = [0.05, 0.1, 0.2]

- Jump distance in between each spectrogram created

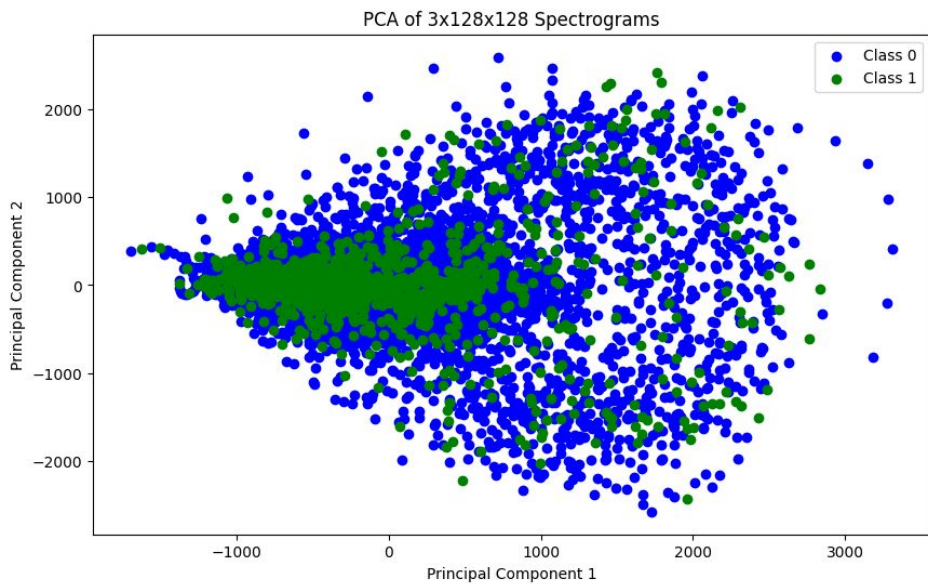


Why is thresh_0.2_step_0.05 50/50?

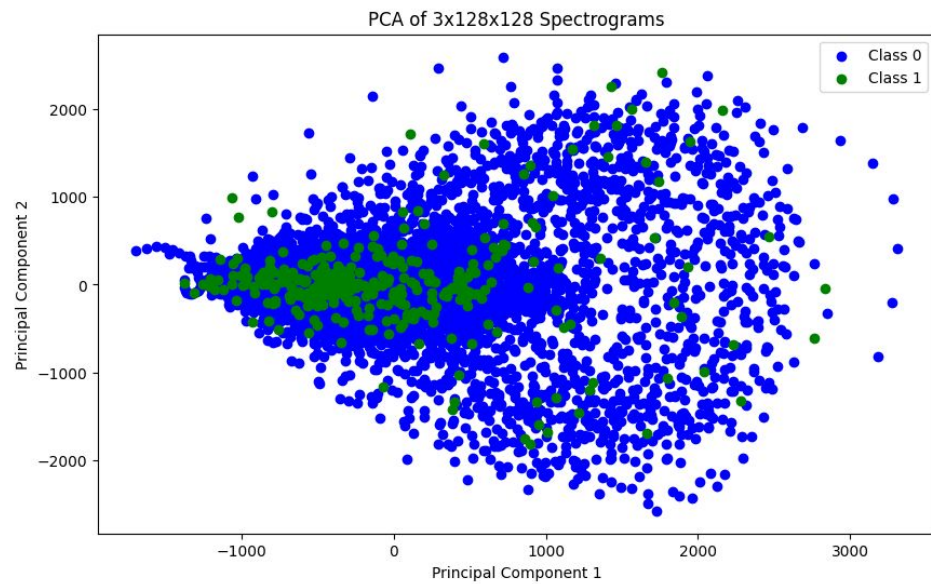


Looking at other distributions

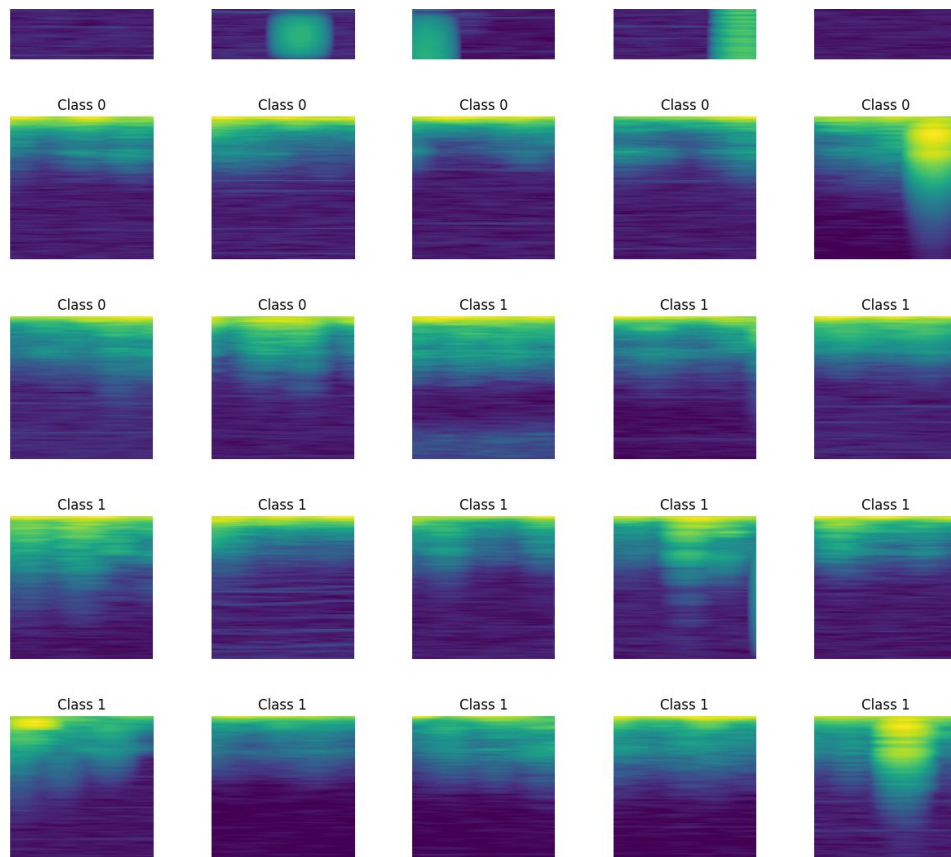
thresh_0.5_step_0.1



thresh_0.7_step_0.1

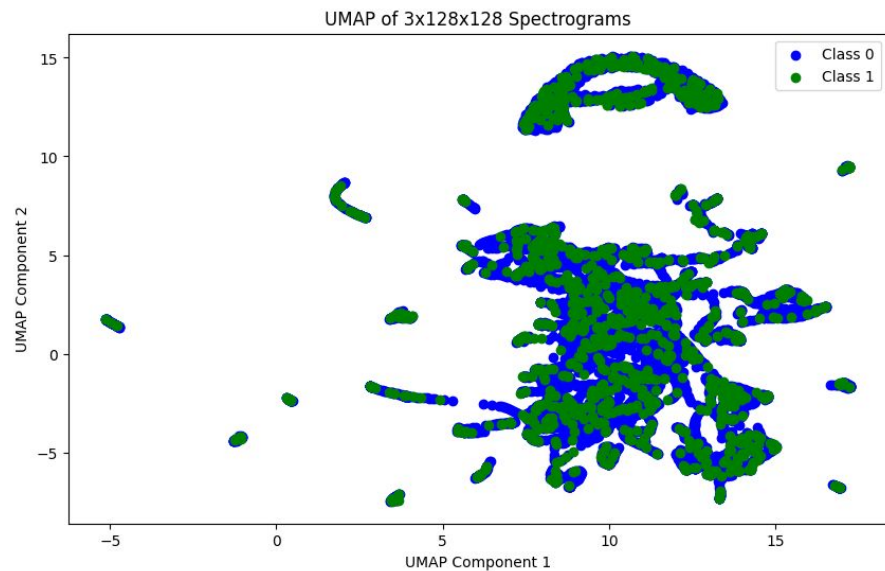
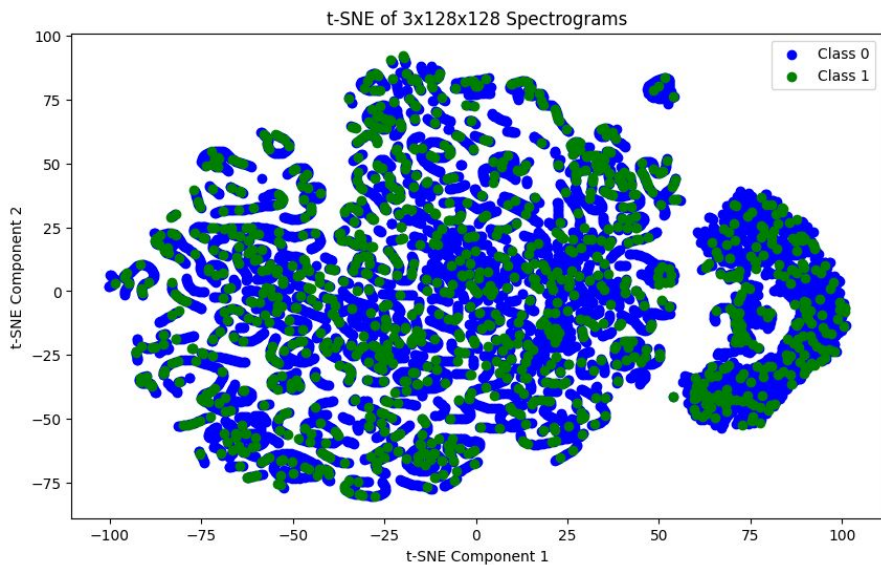


Checking Similar Spectrograms from PCA



Other Comparison techniques on thresh_0.5_step_0.1

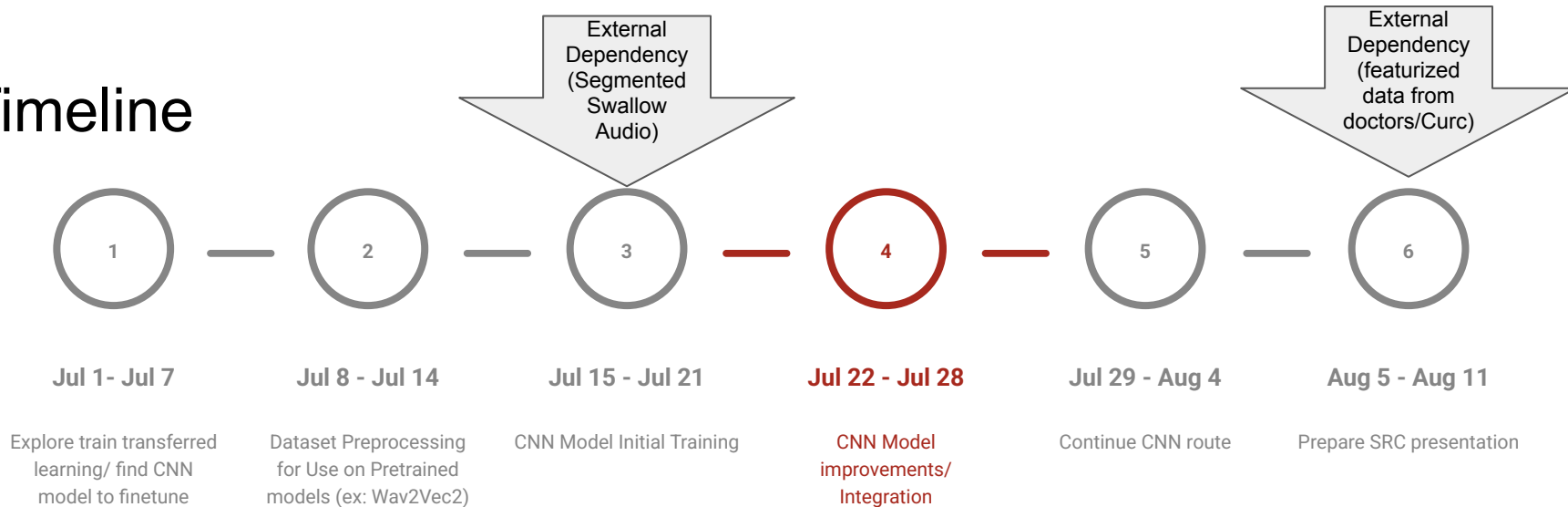
*not familiar with t-SNE or UMAP



Conclusion

- **Improper Data Representation:** The consistent bias towards one class may indicate that the dataset isn't properly assigning labels to swallows and non-swallows. (most likely)
- **Lack of Defining Features:** there may not be enough distinct features to clearly differentiate between the two classes. This could mean that the data lacks the necessary characteristics to separate the classes effectively. (less likely)

Timeline



SRIP Important Dates

JULY 7TH	STUDENT PRESENTER REGISTRATION AND ABSTRACT DEADLINE
STUDENT PRESENTER MODIFICATION DEADLINE	JULY 14TH
AUGUST 9TH	"HOW TO PRESENT AT THE SRC" ZOOM PRESENTATION - 9AM
SUMMER RESEARCH CONFERENCE! PRICE CENTER	AUGUST 14TH & 15TH



Work Remote (only taking 1 course)

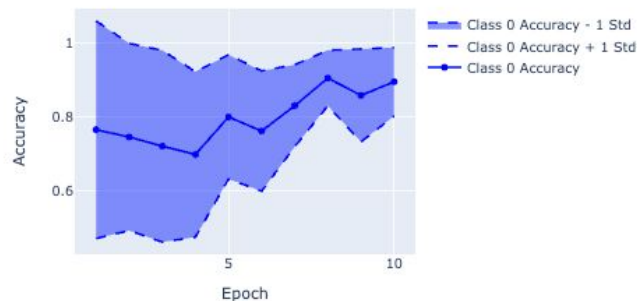
CNN Segmenter Specifics

CURRENT WORK: transitioning to google colab

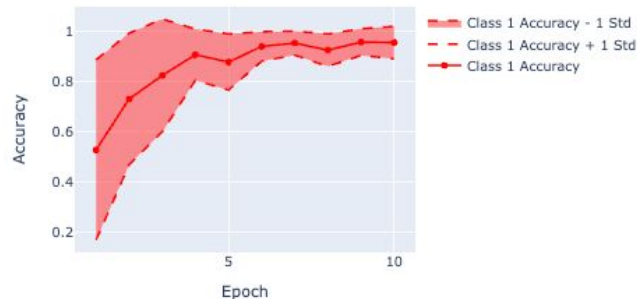
- Not yet tested on the medical group dataset

Validation Accuracy for classes 0,1 over 25 runs of training with Resnet

Validation Accuracy for Class 0



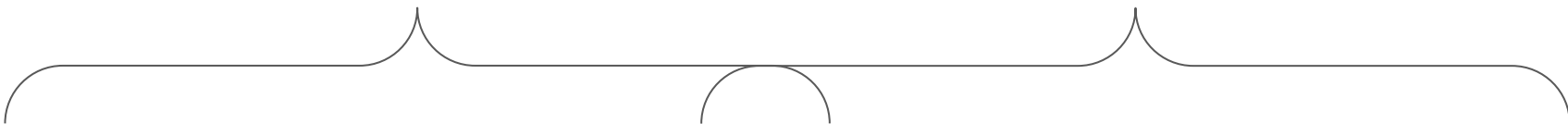
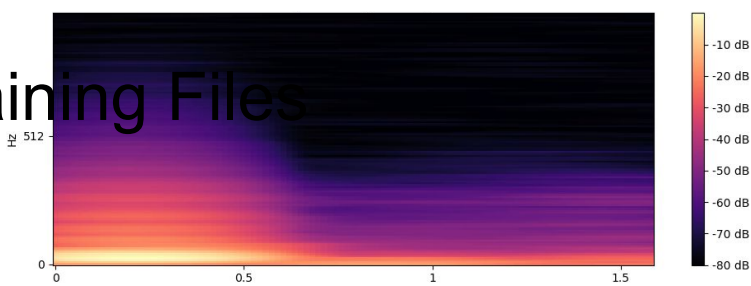
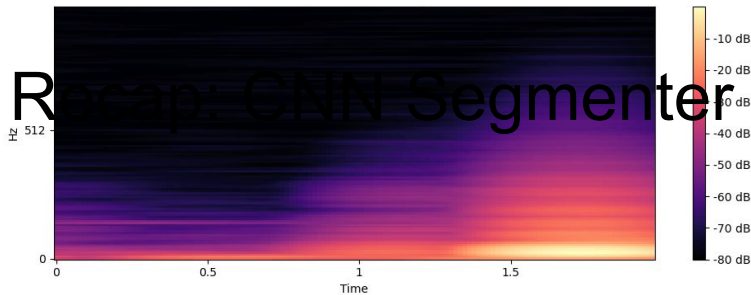
Validation Accuracy for Class 1



TODO

- **Debugging CNN + maybe more**
- Select features of interest and automate
 - Take both the features and code for automation from medical analysis team
- Create Classifier

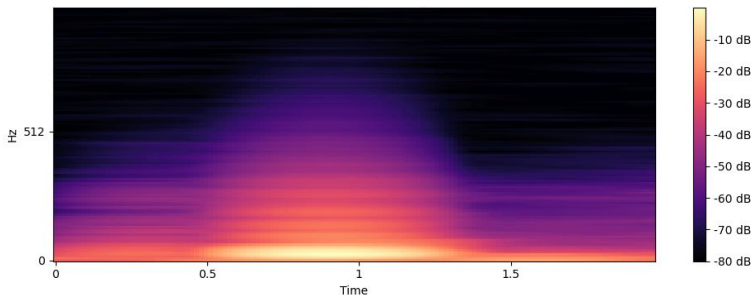
Recap: CNN Segmenter Training Files



0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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4 seconds of audio (100ms timestamps)

Label 1



Issue with accurate spectrogram timestamping relative to the original label vector

Hyperparameters:

1. min # of 1s in segment to label as 1 ($\geq 30\%$)
2. Jump distance (100ms)

Moved to local computer from Google Collab

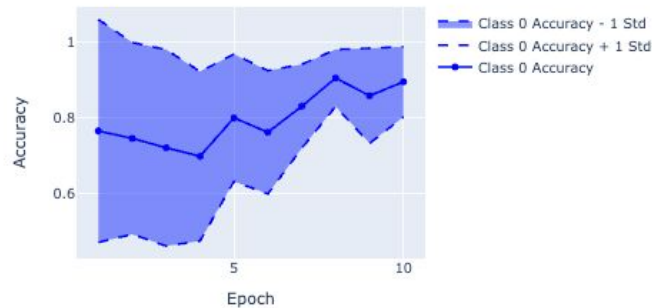
- Change to use mps - Metal Performance Shaders

Compartmentalizing training/testing

- All spectrograms from 1 audio file either in training or in testing.
- No overlap to ensure results are not due to “duplicates in testing”

Previous Accuracy: very high for both classes

Validation Accuracy for Class 0



Validation Accuracy for Class 1

